

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B.Tech. Branch: Mechanical Engg./Mechanical Engg. (Sandwich)/Production

Subject Code & Name: BTMC701 Mechatronics

Semester :VII

Max Marks: 60

Date:-02-01-2024

Duration: 3 Hr.

Instructions to Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.		12
A) Define Mechatronics and list out advantages and disadvantages of mechatronics. Also give some applications	CO1	6
B) Explain with neat sketch working principle of strain gauges. How it can be used as pressure transducer?	CO1	6
C) Explain electromagnetic frequency domain transducer with neat sketch.	CO1	6
Q.2 Solve Any Two of the following.		12
A) Explain Digital signal with its advantages, disadvantages, and applications.	CO2	6
B) Explain the need for Signal processing.	CO2	6
C) Describe the operational amplifier in detail.	CO2	6
Q. 3 Solve Any Two of the following.		12
A) Explain working principle of stepper motor with neat sketch. Give its classification	CO3	6
B) What are the various types of pressure control valves used in hydraulic systems? Explain any one with neat sketch	CO3	6
C) Explain as to what do you mean by pneumatic air preparation and significance of FRL unit.	CO3	6
Q.4 Solve Any Two of the following.		12
A) Draw and explain the architecture of the 8085 microprocessor.	CO4	6
B) What are the addressing modes of the 8085 microprocessor?	CO5	6
C) What do you mean by microcontroller? Differentiate between microcontroller and microprocessor.	CO5	6
Q. 5 Solve Any Two of the following.		12

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| A) Differentiate between Open loop system and closed loop system giving suitable examples. | CO5 | 6 |
| B) Define transfer function. Explain procedure to determine transfer function of a control system. | CO5 | 6 |
| C) What is a PID Controller? How Does a PID Controller Work? | CO6 | 6 |

***** End *****